Hemodynamic Role of Extracellular Fluid in Hypertension

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SUMMARY **Plasma volume is usually lower in patients with essential hypertension than in normal subjects. Normal or expanded plasma volume in some hypertensive patients may respect either a specific hypervolemic subset of the disease or the upper end of a continuum of volume values. Difficulties in defining these groups stem from the small numbers of subjects studied, from the need for a reliable reference index for volume measurements, and from the multiple factors which may affect intravascular volume. Differences in plasma volume cap influence choice of antihypertensive therapy; patients with expanded volume tend also to have slightly higher exchangeable sodium and greater extracellular fluid (ECF) volume and to respond well to diuretic therapy. There is also some evidence that low plasma renin activity is more frequent among hypervolemic patients. Essential hypertensives as a group have low plasma to interstitial fluid volume ratio (PV/IF), indicating that ECF distribution between the intravascular and interstitial compartments is shifted toward the latter. This is probably related to altered capillary filtration pressure due to increased venous resistance. Hypovolemic essential hypertensives have significantly lower ($P < 0.01$) PV/IF ratio than hypervolemic, but whether this is related to differences in neural venous tone is only speculative. Hemodynamic studies revealed no difference in cardiac output between hypertensive patients with contracted blood volume and those with hypervolemia; total peripheral resistance was even higher in the latter, suggesting that "vasoconstriction" is not different between the two groups. It is widely believed that the relationship between ECF expansion and hypertension depends on the development of hypervolemia, increased cardiac output, and subsequent rise in total peripheral resistance reducing volume expansion and normalizing systemic flow while maintaining a high blood pressure. This sequence of events has been demonstrated in some human and experimental forms of hypertension but not in all. Metyrapone-induced hypertension in dogs could be sustained for up to 6 weeks by increased output with no evidence of "autoregulation" developing, and similar observations were reported in some anephric patients. Complementing these findings are observations of elevated cardiac output in some patients with long-standing essential hypertension or primary aldosteronism. It is therefore suggested that the spectrum of hemodynamic changes associated with volume disturbances in hypertension is too wide to be forced under one hypothesis alone and that neurogenic and other factors may play an important role in that complex relationship.**

THE IMPORTANCE of extracellular fluid (ECF) volume variations in hypertension hardly needs stressing; they have been described at every stage of the hypertensive process. At the onset of experimental hypertension, ECF expansion has been reported not only with the rise in blood pressure due to salt loading[1-3] or steroid administration,[4,5] but also in renovascular hypertension.[6,7] Various disturbances in plasma and ECF volume and especially of their relationship to arterial pressure levels have been described in different types of established hypertension[8-12] and have revealed subgroups with increased or decreased plasma volume within the ill-defined entity of "essential" hypertension.[9,13] More recently, essential hypertension has been analyzed in terms of "volume" and "vasoconstriction," the former assumedly due to increased flow and arterial overfilling.[14] Finally, control of hypertension is associated with, and markedly influenced by, alterations in ECF volume.[10,15-17]

What is less clear, however, is the mechanism by which ECF volume and arterial hypertension are linked. Beginning with Starling[18] and later Borst,[19] a well structured concept gradually evolved based on the work of many investigators, especially Ledingham,[3] Guyton and Coleman[20,21] and their collaborators. Increased ECF and blood volume resulting from renal disease or dysfunction, excess salt intake, or increased steroids lead to a high cardiac output and hypertension. Eventually autoregulatory processes return cardiac output to normal and hypertension is maintained by increased total peripheral resistance (TPR). Classic examples of this progression can be seen in many hemodynamic studies of various types of experimental renal hypertension[1-3,6,7,22-24] and in the observations of Coleman et al.[25] in nephrectomized patients. Licorice-induced hypertension also appears to follow the same pattern.[19] However, this sequence of events from increased ECF to increased TPR through hypervolemia and a rise in cardiac output is not always seen either experimentally[26-28] or clinically.[29-30] Luetscher et al.[31] have also suggested that autonomic neural influences were capable of adjusting to the excesses of fluid reported without necessarily leading to hypertension.

The following review is, therefore, a discussion mainly of human studies in order to assess the incidence and type of ECF volume abnormalities in hypertension and their hemodynamic correlates. Especially stressed will be the difficulties encountered in relating clinical observations to some current pathophysiological concepts. The aim is to outline the many areas where further work and studies are needed and the need for a multifactorial approach even when one aspect such as volume is investigated.

Materials and Methods

In addition to previously published data, this review will refer to information derived from a larger number of patients with untreated essential hypertension and including determination of exchangeable sodium (Na_e), plasma renin activity, (PRA), and cardiac output in addition to ECF, plasma volume, and cardiopulmonary volume (CPV). The methods used have been described in detail previously;[32] plasma

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volume was measured in the morning after the subjects had had at least ½ hour of supine rest, using ^{125}I-human serum albumin (HSA) and a 10-minute equilibration period. Total blood volume (TBV) was calculated from plasma volume and simultaneously determined hematocrit with appropriate correction factor for the difference between total body and large vessel hematocrit. PRA was estimated by radioimmunoassay of angiotensin I generated during incubation of 1.0 ml of plasma for 3 hours at pH 5.7;[33] the plasma was obtained from blood samples drawn immediately before blood volume determination.

Cardiac output and CPV were determined immediately following the plasma volume measurement; appropriate catheters were introduced percutaneously and carefully positioned in the ascending aorta and at the junction of the superior vena cava with the right atrium. Classic formulas based on the Stewart-Hamilton technique were used to calculate cardiac output and CPV from dye-dilution curves (indocyanine green).[32] ECF volume was determined with ^{81}Br, using an 18-hour equilibration period from the afternoon before the morning on which the plasma volume was determined.[34] In other subjects, total Na_e was determined using 30 μCi of ^{24}Na and a 24-hour equilibration time according to the method described by Moore et al.[35] All patients had been thoroughly investigated and had either had no treatment or had discontinued all antihypertensive drugs at least 1 month before the studies. None had a history or evidence of cardiac or renal decompensation or malignant hypertension.

ECF Volume in Essential Hypertension

Most investigators have reported normal values for ECF volume in essential hypertension;[11, 32, 36-38] studies reporting increased ECF had included an appreciable number of patients with renal parenchymal disease,[39] with cardiac impairment,[40] or with increased Na_e and advanced fundal changes[41] ECF expansion is the rule in cardiac decompensation even in the absence of obvious edema.[11] The combination of hypertensive retinopathy, expanded ECF, and increased Na_e reported by Ross[41] corresponds to the findings by Hollander et al.[11] in malignant hypertension. Similar to these findings in malignant hypertension, and in contrast with essential hypertension, the increased blood pressure produced in man by primary aldosteronism is associated with expanded ECF.[12, 37, 42] Braun-Menendez and Martinez[4] in much earlier studies showed that deoxycorticosterone acetate led to increased ECF in rats. These observations should not be construed to mean that all "steroid hypertensions" are necessarily associated with volume expansion; not all patients with primary aldosteronism have increased ECF (Tarazi, Bravo, and Dustan, unpublished observations) or plasma volume.[43] Walser et al.[44] stressed the differences in volume effects between cortisol and deoxycorticosterone. Our own studies in dogs made hypertensive with metyrapone (100 mg/kg per day) have shown that ECF expansion does not necessarily occur with the rise in blood pressure.[45]

In a study of suppressed PRA in essential hypertension, Jose et al.[46] found that six patients with low PRA had an expanded ECF ($P < 0.01$) and larger plasma volume (1.53 vs. 1.28 liters/m^2, $P > 0.05$) than essential hypertensives with normally responsive PRA. Distler et al.[47] also recently reported that hypertensive patients with low plasma renin concentration (1.14 ng/ml supine and 1.81 standing vs. 4.32 and 10.69 ng/ml, respectively, for "normal renin" hypertensives, $P < 0.001$) had higher plasma volume than patients with normal renin concentration (1.71 vs. 1.57 liters/m^2, $P < 0.025$). On the other hand, Schalekamp et al.[48] could not document a significant difference in either plasma or ECF volume between hypertensive patients with high or low PRA; however, their patients were investigated while on a fixed, relatively low (50 mEq/day) sodium intake for 5-7 days. The discrepancy between their results and the others could, therefore, be due in part to the tendency to greater volume loss with sodium depletion in patients with expanded ECF and low PRA.[46] This suggestion is supported by unpublished observations from Mimran, Fourcade, and Barjon, who found that significant difference in plasma volume among their untreated hypertensive patients disappeared following low sodium diets.

Plasma Volume in Essential Hypertension

Studies of the intravascular component of ECF have shown a wide spectrum of values in hypertension[9, 11, 46, 49] (see Tarazi et al.[9] and Dustan et al.[10] for references to previous work). We accordingly suggested that essential hypertension could be differentiated into subgroups: "hypovolemic and hypervolemic" types. This subdivision was based on the relationship of plasma volume to diastolic pressure levels; however, similar subdivisions based on "volume factors" were also suggested by other investigators using different methods of approach. Jose et al.[46] reported increased ECF volume in some essential hypertensives; Woods et al.[50] described differences in Na_e in uncomplicated essential hypertension. Vaughan et al.[51] suggested that hypertensives with low plasma renin responded preferentially to diuretic therapy and therefore probably have a "high volume type" of hypertension. It is tempting to suppose that these different studies were outlining from different points of view the same subtype of hypertension. Unfortunately, however, available data do not allow a conclusive answer.

Since a correlation between volume expansion and suppressed renin activity has often been demonstrated,[52-54] the suggestion that low renin hypertension was associated with hypervolemia seemed a logical assumption.[14, 51] However, reports describing simultaneous determinations of plasma and ECF volumes, Na_e and PRA have been relatively few and some gave conflicting results. Three groups of investigators[46, 47, 49] reported volume expansion in low renin patients, although one of them[47] found no significant difference in Na_e among patients with different plasma renin concentration. On the other hand, the Glasgow group reported that low renin hypertension—except for primary aldosteronism—was not associated with volume expansion because they found no difference in plasma or ECF volume[48] or in Na_e[55] between hypertensives with high or low PRA. Their patients, however, were investigated during moderate sodium restriction which might have blunted volume differences in group averages.

Whatever the reason for these discrepancies, their occurrence is really not quite unexpected because of the number of factors that can influence either PRA or plasma volume. In two separate studies of man with essential hypertension, only 25% of variations in PRA were determined by variations in plasma volume ($r = -0.502$).[34, 52] Thus, the lack of significant differences (among a particular group of subjects) between "high and low renin" patients does not necessarily discount the possibility of a hypervolemic subgroup. Furthermore, it must be recalled that differences, at least as far as plasma volume is concerned, were described in relation to other hemodynamic abnormalities[9, 10, 56] and that lumping together all essential hypertensives might obscure important variations.

The basic consideration is whether differences in plasma volume among patients constitute distinct subsets of essential hypertension as has been proposed.[10] In the spectrum of volume values found among essential hypertensives, the hyper- and hypovolemic types may only represent extreme examples of a wider distribution curve. A full discussion of that question (see Tarazi et al.[9] and Dustan et al.[10]) is outside the scope of this review, which proposes mainly to investigate the hemodynamic impact of differences in intravascular volume. However, the terms by which "volume" is defined must be clarified and only the main lines of the argument (which is still open) can be sketched.

An important characteristic of the majority of essential hypertensives is a reduction in plasma volume; the evidence for this decrease in plasma volume has been fully discussed in previous papers.[8-10] In addition, we have repeatedly found significant relationships between intravascular volume and arterial pressure. Studies of essential hypertension showed an inverse correlation between the weekly average diastolic pressure and plasma volume, contrasting with the positive correlation seen in patients with parenchymal renal disease.[9, 29, 57] The difference was in the sign of the correlation, not necessarily in the magnitude of the volumes involved; it stressed the need to relate volume findings in hypertension to the hemodynamic abnormality that defines the disease, namely, the level of the arterial pressure. This inverse correlation between plasma volume and arterial pressure was confirmed by Julius et al.[58] by Parving and Gyntelberg[59] and by Ulrych,[60] and indirectly by Bello et al.[61] However, the correlation of pressure with volume appeared restricted to men only;[9] in contrast, the correlation between blood volume and TPR was more frequently significant because it was obtained in both men and women.[56] This inverse relationship was confirmed by many centers and in different types of hypertension; it was documented in essential hypertension,[56, 61, 62] in borderline hypertension,[58, 63, 64] in renal arterial disease,[56] and in primary aldosteronism.[43] It was also described in normotensive subjects by Julius et al.[58] and by Safar et al.[63, 65]

Against this background the description of hypervolemic essential hypertension became possible.[9, 66, 67] A small group of patients was first recognized in 1970[9] because of the marked contrast between their plasma volume and the low values generally found in essential hypertensives with equally elevated diastolic pressure (> 110 mm Hg, weekly average) (Fig. 1).[9] Their hypertension was not related to any obvious cause, they had little if any hypokalemia and no recognized steroid abnormality, and showed a remarkable depressor

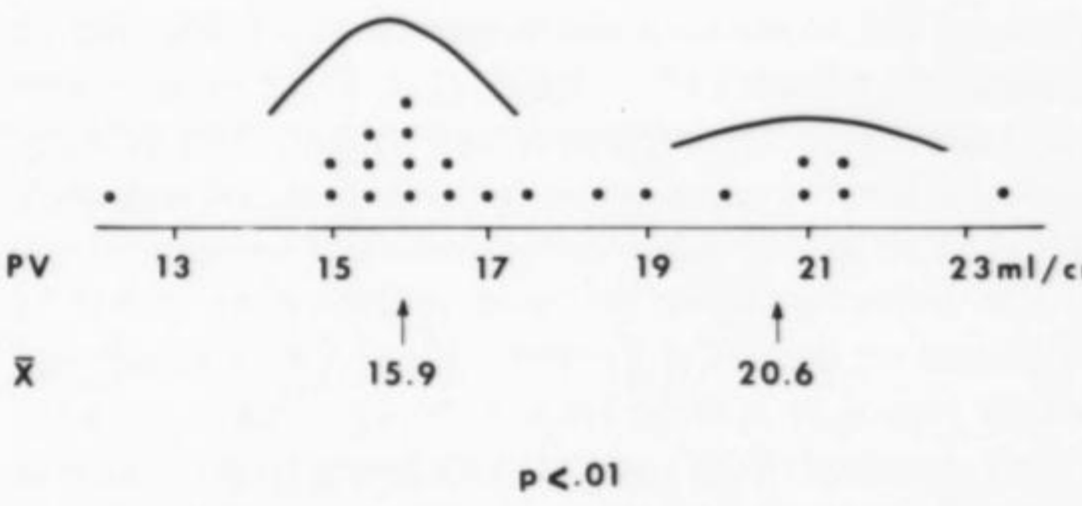

FIGURE 1 *Individual values for plasma volume (PV) in 22 men with marked essential hypertension (described in 1970[9]). Average for total group (17.3 ± 0.53 ml/cm) was significantly (P < 0.05) lower than normal (18.7 ± 0.33) but patients appeared to fall into two groups with significantly different plasma volumes (15.9 ± 0.37 in 14 patients vs. 20.6 ± 0.70 in eight).*

response to diuretic therapy.[9, 67] With continued experience, the number of these patients increased although they remained a distinct minority as compared to essential hypertensives with contracted plasma volume.

The latter consistently demonstrated an inverse correlation between blood volume and TPR ($r = -0.446$, $P < 0.025$) (Fig. 2) similar to that described in essential hypertensives and in normal subjects from other centers.[58, 63-65] The correlation between total blood volume and TPR in our small group of normal volunteers was not statistically significant, but the values obtained seemed to fall in line with those of hypovolemic hypertensives. Adding the two

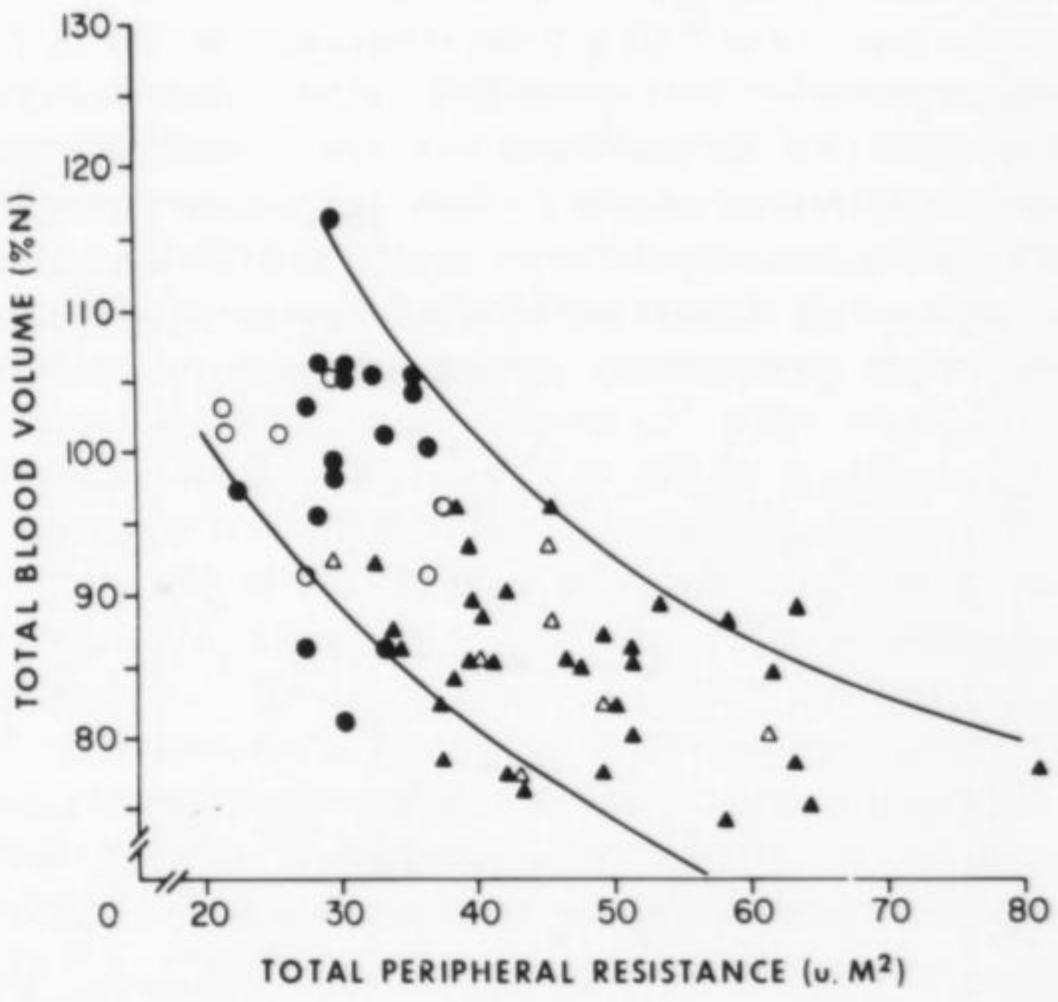

FIGURE 2 *Relation between total peripheral resistance and total blood volume in 25 normotensive subjects and 38 patients with essential hypertension associated with normal or contracted plasma volume. Total blood volume is expressed as percentage of normal to allow inclusion of both men and women despite sex differences in intravascular volume.[9, 56] Total peripheral resistance is expressed in arbitrary units corrected for body surface area. The correlation coefficients for patients with essential hypertension was significant (r = −0.445, P < 0.02); data from the normotensive subjects fitted well with those from the hypertensives and increased the correlation coefficient for the whole group (r = −0.603, P < 0.001). Circles represent normotensive men (●) and women (○); triangles represent hypertensive subjects, men (▲) and women (△).*

increased the correlation coefficient ($r = -0.603$ and its statistical significance ($P < 0.001$) (Fig. 2). In contrast with both hypovolemic hypertensives and patients with primary aldosteronism,[43] hypervolemic hypertensives did not show an inverse relationship between volume and resistance (Fig. 3). The distinction based on these considerations could be questioned on statistical grounds. Viewed as a group and without regard to clinical criteria, background experience, or slow evolution of the concept, hypertensive patients could be viewed as showing a continuum of plasma volume values with wide variations and no relationship to TPR. This mathematically valid viewpoint would not take into account the expected reduction of plasma volume at high levels of TPR reported by different investigators.[58, 62-65] The argument between a qualitative biological interpretation and a quantitative statistical outlook remains open. This, however, does not detract, in our opinion, from the possibility of comparing the hemodynamic indices of patients derived from both ends of the spectrum.

To investigate whether these differences in plasma volume were also associated with differences in ECF, the latter was determined in a group of 31 patients with essential hypertension. Results are summarized in Table 1 and compared with a similar study by Ibsen and Leth.[38] Averages for the whole group showed no difference in ECF volume between normal subjects and patients with essential hypertension. However, separation of essential hypertensives on the basis of plasma volume into hypo- and hypervolemic groups revealed a different picture (Table 2). Values for normotensive men in our laboratory averaged 1.71 ± 0.03 (SEM) liters/m^2 for plasma volume[9] and 9.10 ± 0.026 (SEM) liters/m^2 for ECF. When expressed in relation to body weight, there was no difference in ECF between hypo- and hypervolemic patients; however, expressing results in liters per square meter of body surface area showed an expansion in ECF similar to that reported by Jose et al.[46] Calculations in reference to body height gave similar results: 99 ± 3 ml/cm for

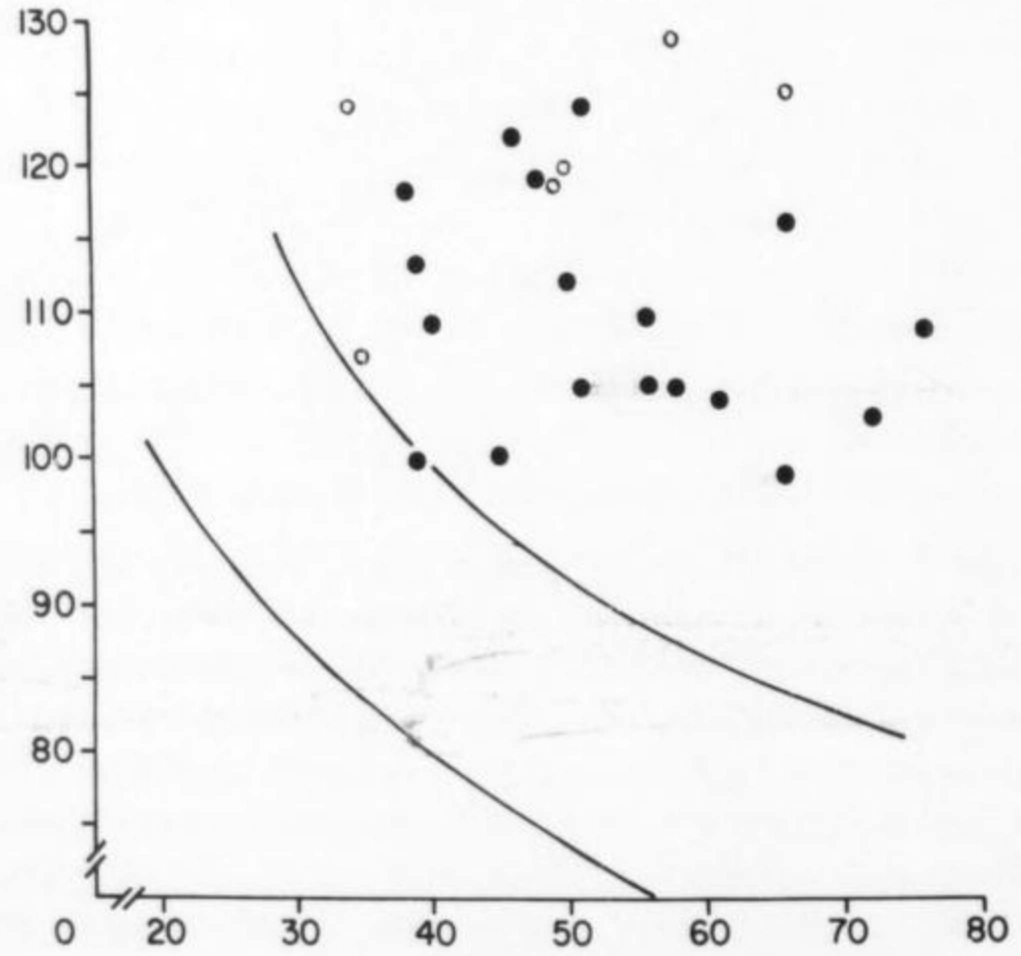

FIGURE 3 *Hypervolemia with essential hypertension is best defined in relation to level of total peripheral resistance; values for total blood volume in these 24 patients [18 men (●) and six women (○)] fall outside limits of relationship defined in Figure 2.*

normotensives and 102 ± 2.7 ml/cm for the hypovolemic hypertensives, as opposed to 120 ± 2.7 ml/cm for the hypervolemic group ($P < 0.01$). Similar variations in interpretation, depending on the mode of expression of results, were reported by Novack et al.;[37] they found that ECF in primary aldosteronism was expanded when expressed in percent of body weight but "normal" when expressed in relation to total body water. Similarly, the observations of Distler et al.[47] showed that plasma volume expressed in liters per square meter was higher in low renin hypertensives than in normal renin patients ($P < 0.025$), but this volume difference disappeared if blood volume was expressed in milliliters per kilogram (41.8 vs. 41.4 ml/kg).

These observations stress the need for, and the difficulty of, carefully selecting an appropriate index of reference to express body fluid volumes. The undue influence of body weight because of its relatively water-poor fat component, on expressions of body fluid volume has been repeatedly stressed.[9, 35, 68] Increased body weight in adults usually implies a fatter individual; therefore virtually all aqueous-phase components of body composition, if expressed as simple ratios of body weight, will be found to decrease with rising body weight.[35] This observation detracts from the value of reports expressed exclusively in milliliters per kilogram or as percent of body weight. Reference to body height or to some expression involving body height[69] can help overcome to some extent this difficulty.[8, 68, 69]

Similar findings were obtained in a study of Na_e in another group of patients, 14 normal male volunteers and 16 men with essential hypertension (Table 3). The average for all hypertensive patients (39.3 mEq/kg) was not significantly different from normal; the difference between the hyper- and hypovolemic subgroups of hypertensives varied both in sign and in significance with the index of reference (weight or body surface area) used. Our experience has been that many hypervolemic essential hypertensives are markedly overweight, even compared to other hypertensives as shown by the averages in Table 2 and 3. This difference in weight makes data interpretation more difficult, but our results, like those of Woods et al.,[50] suggest the presence of subtle differences in Na_e between subsets of patients with uncomplicated essential hypertension. However, neither Lebel et al.[55] nor Distler et al.[47] found significant differences in Na_e between hypertensive patients with different plasma renin levels.

ECF Partition in Hypertension

The finding of reduced plasma volume and normal ECF suggests a disturbance in the partition between the intravascular and interstitial components of ECF. Both Tarazi et al.[34] and Ibsen and Leth[38] found significant reduction of plasma to interstitial fluid volume (PV/IF) ratio in men with essential hypertension; this was confirmed in a larger number of patients (Table 1). Since the relation of plasma volume to ECF is particularly stable in normal subjects,[70] the alteration of PV/IF in hypertensive patients suggests some disturbance in the forces regulating ECF partition. A more direct demonstration of increased capillary filtration rate in essential hypertension was obtained by Parving and Gyntelberg[59] and by Ulrych.[60] The transcapillary escape

TABLE 1 *Extracellular Fluid (ECF) and Plasma Volume (PV) in Men with Essential Hypertension (EH)*

Series	Subjects (no.)	PV (ml/M² ± SE)	ECF (liters/M² ± SE)	PV/IF (av ± SE)
Present*	Normals (11)	1652 ± 38	9.10 ± 0.026	0.223 ± 0.003
	EH (31)	1481 ± 30	9.32 ± 0.137	0.190 ± 0.004
	P	<0.001	NS	<0.001
Ibsen and Leth[38]	Normals (18)	1891 ± 44	9.24 ± 0.19	0.257 ± 0.01
	EH (18)	1682 ± 31	9.21 ± 0.18	0.223 ± 0.006
	P	<0.001	NS	<0.005

* Includes data from 11 normals and 20 EH subjects reported previously (Tarazi et al.[34]).
PV/IF = plasma to interstitial fluid volume ratio.

TABLE 2 *Hypo- and Hypervolemic Essential Hypertension: Extracellular Fluid (ECF) Characteristics in Two Groups of Men with Essential Hypertension*

	Hypovolemic (20 pts)	Hypervolemic (11 pts)	*P*
Blood pressure (mm Hg) (week average)	167/108	160/105	NS
Weight (kg)	87.6 ± 2.4	97.0 ± 4.8	<0.01
Plasma volume			
ml/cm	15.8 ± 0.28	20.2 ± 0.46	<0.001
ml/M²	1401 ± 22	1702 ± 50	<0.001
ECF			
% bw	22.2 ± 0.41	22.5 ± 0.92	NS
liters/M²	9.09 ± 0.18	10.02 ± 0.25	<0.01
PV/IF	0.184 ± 0.004	0.206 ± 0.008	<0.01

bw = body weight; other observations as in Table 1. Values given are average ± 1 SEM.

rate of albumin in the capillary level (Fig. 4) was directly correlated with mean arterial pressure; this could be assumed to reflect an elevated capillary filtering pressure resulting either from inadequate protection by precapillary sphincters and arterioles against the high systemic pressure or from increased venular tone.[10, 59] Our findings of an inverse relationship between intravascular volume and TPR[56] might be taken to support the former possibility, since modulation of capillary pressure might be less precise with increasing arterial pressure.[59]

It is more likely, however, that the reduced PV/IF ratio is related to increased venous resistance.[8, 34, 71] This is suggested by the essentially normal values of cardiac output and right atrial pressure in most of the essential hypertensives under consideration. In possible support of this explanation is the finding of decreased venous distensibility in hypertensive patients.[72] Further, we have found an increased central blood volume to total blood volume ratio in some hypertensive subjects with reduced plasma volume, indicating a decreased reservoir function of large veins;[32, 73] this could be associated with slight increases in venular tone. Studies in dogs with renal hypertension have shown an increased mean circulatory pressure,[27] which implicates the venous side of the circulation as well as the arterial in the maintenance of hypertension. Increased pressure at the venular end of the capillary bed could even lead to vasoconstriction as has been suggested by studies of isolated vascular beds.[7] Recently, Leth and Ibsen[71] reported that the PV/IF ratio was further diminished by propranolol therapy, suggesting venoconstriction as the more probable explanation. Previous studies with guanethidine have also demonstrated that neural control of veins could play an important

TABLE 3 *Exchangeable sodium (Na_e) in 16 Men with Essential Hypertension*

Group and no. of patients	Weight (kg + SE)	Exchangeable sodium (mEq/kg ± SE)	Exchangeable sodium (mEq/m² ± SE)	
1. Hypovolemic (9)	73.9 ± 3.6	41.3 ± 1.23	1641 ± 36.7	NS
2. Hypervolemic (7)	104.2 ± 4.8	36.9 ± 0.90	1736 ± 34.4	<0.01
P (1 vs. 2)	<0.001	<0.02	>0.05	
Normotensive (14)	79.7 ± 3.2	38.8 ± 1.02	1576 ± 38.7	

A significant inverse correlation was found between Na_e (mEq/kg) and body weight for all subjects investigated ($r = -0.637$, $P < 0.005$).

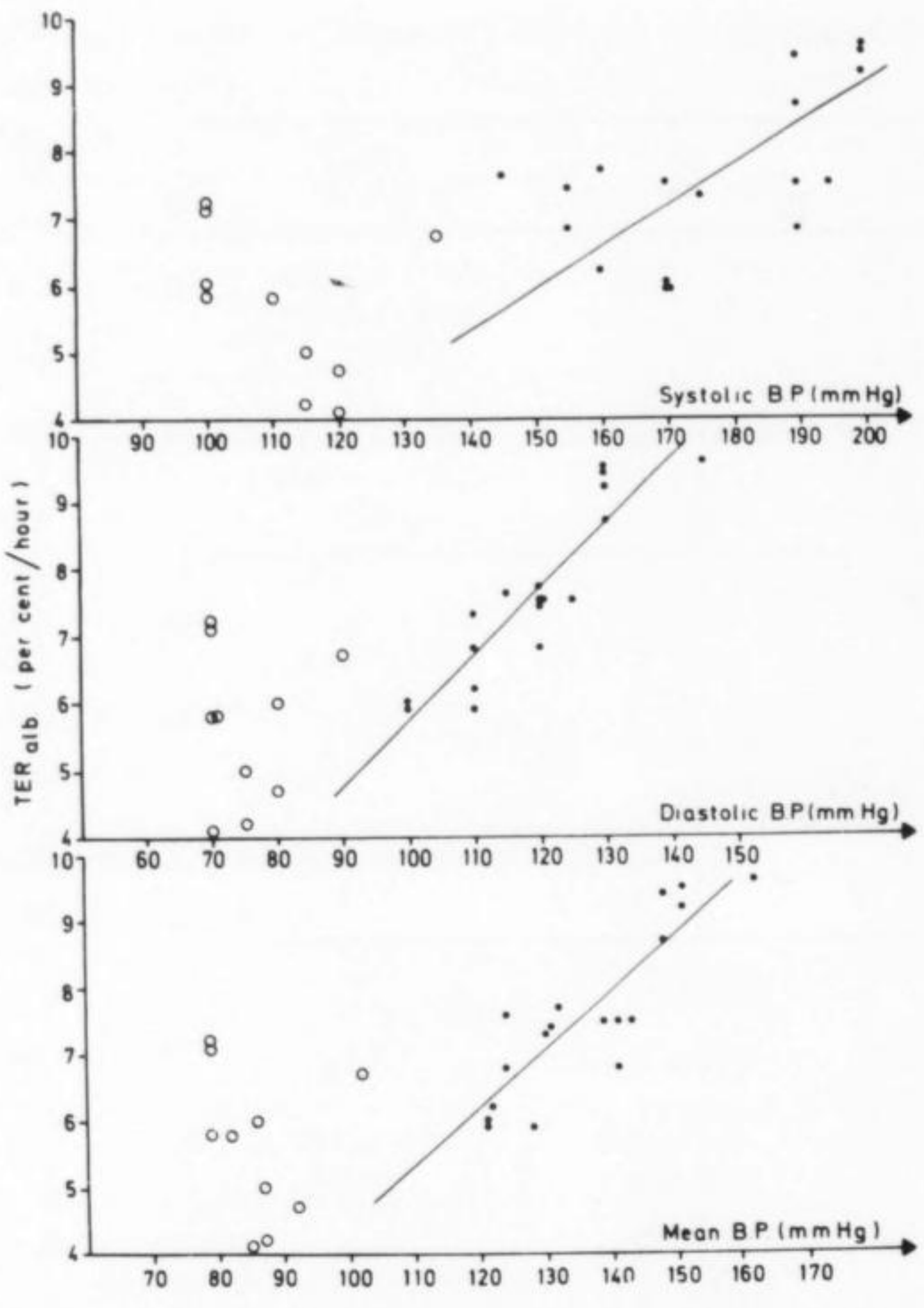

FIGURE 4 *Correlation between transcapillary escape rate of albumin* (TER_{alb}) *and systolic* ($y = 0.061x - 3.24$, $r = 0.061$, $P < 0.01$), *diastolic* ($y = 0.098x - 4.05$, $r = 0.90$; $P < 0.001$), *and mean blood pressure* (*B.P.*) ($y = 0.088x - 4.34$, $r = 0.87$, $P < 0.001$) *in the hypertensive group* (*solid circles*). *No correlation existed within the normotensive group* (*open circles*). *From Parving and Gyntelberg,*[59] *with permission.*

role in regulating volume distribution within the ECF.[75] If this is the main mechanism responsible for the reduced PV/IF ratio in essential hypertension, the results outlined in Table 2 suggest less pronounced venoconstriction in hypervolemic patients. Hypertensives with contracted plasma volume had a significantly lower PV/IF ratio than both normal subjects ($P < 0.001$) and hypervolemic hypertensives ($P < 0.01$). Although the latter also had a slightly reduced ratio (0.206 ± 0.008), this was not significantly different from normal ($0.05 > P < 0.10$).

However, not all types of hypertension present the same type of disturbance in volume distribution. In contrast with the reduced PV/IF ratio found in essential hypertension, the partition of ECF between its intravascular and interstitial compartments is skewed toward the former in rats with renal hypertension.[76, 77] Lucas and Floyer[76] have shown that the PV/IF ratio is increased in both renovascular and renoprival hypertensive rats. Despite an equal ECF expansion in bilateral nephrectomized rats and in rats with ureterocaval anastomosis, hypertension developed only in the former, and the rise in blood pressure was associated with selective expansion of intravascular volume. They suggested, therefore, that the kidneys influence the compliance characteristics of the interstitial gel, allowing it to "store" more water with less rise in pressure.

Thus, both neural and possibly humoral (renal) factors can alter the distribution of fluid volume between the vascular system and the interstitial space, which may help explain some of the variations in the relationship of ECF with arterial pressure.

Intravascular Volume and Cardiac Output

Chronic volume expansion or depletion need not necessarily lead to parallel changes in cardiac output. Although a sudden increase in blood volume causes an immediate and sometimes marked increase in cardiac output, a slowly developing hypervolemia of equal magnitude does not cause a significant change in cardiac output.[78] This is due to the marked adaptability of the venous system in which both finely tuned neural controls[79] and stress relaxation[78] play important roles. Venous return seems related more to mean circulatory pressure than to blood volume,[78, 80, 81] and the level of cardiac output will therefore depend more on the redistribution of blood volume between the peripheral and cardiopulmonary segments than on the magnitude of volemia.[73] Another important factor under certain conditions is the level of cardiac performance.[78, 79]

Ferrario et al.[27] have reported increased cardiac output in the early stages of perinephritic hypertension in the absence of plasma volume expansion. In fact, plasma volume was lower in the face of increased output and rising mean circulatory pressure, suggesting venoconstriction.[28, 32] Similar conclusions were reached from the study of CPV in established[32, 73] and borderline[32, 63, 73, 80] essential hypertension. No relationship was found between volume and cardiac output in either normal[58] or hypertensive[53, 73] subjects. Reports from different centers have demonstrated that the high output of borderline hypertension was not dependent on hypervolemia.[32, 63, 73, 83, 84] In established hypertension, we could find no difference in cardiac output between hypo- and hypervolemic groups despite a marked difference in blood volume (Table 4). In contrast, a significant direct correlation was found between cardiac output and CPV or between cardiac output and the fraction of intravascular volume present in the central circulation (CPV/TBV ratio).[32, 73] This correlation was documented by many investigators both in hypertension and in other conditions,[85, 86] but not by all.[87] Although both cardiac output and CPV are calculated from the same curve, Milnor et al.[85] and others[87-89] have shown that the correlation between these two indices was not necessarily artificial. It was not invariably found in all conditions, and more importantly cardiac output and CPV may be altered independently of each other.[32, 83, 87] The demonstrable dependence of that relationship on neurogenic factors[32, 83] may help explain the diversity of results regarding its statistical significance.

Among patients with widely diverging levels of sympathetic activity such as essential hypertensives,[90-92] it is little wonder that the cardiac output-CPV volume relationship is not of a high order. In contrast, better definition of the groups investigated reveals the correlation[63] or increases its coefficient;[32] the same result is obtained by cardiac autonomic blockade[83] or, contrariwise, by administration of isoproterenol.[32] Both maneuvers tend to reduce differences in sympathetic drive among patients, one by dampening the

TABLE 4 *Relation of Hemodynamic Functions to Total Blood Volume in Essential Hypertension (86 Patients)**

Group (no.)	TBV % SE	HR	DBP (mm Hg) week av	DBP (mm Hg) study	Cardiac index (ml/min M²)	TPR (U/M²)	Cardiopulmonary volume† ml/M²	Cardiopulmonary volume† CPV/TBV
Hypovolemic (58)	84.6	78	101	99	2925	43	550	24.3
SE	0.86	1.4	1.9	2.1	88	1.6	24	1.16
Hypervolemic (28)	110.1	72	110	117	2939	50	649	23.4
SE	1.87	2.9	2.9	4.0	105	2.3	21	1.15
P	<0.001	NS	<0.025	<0.001	NS	<0.025	<0.05	NS

* Including men and women with borderline or established essential hypertension.
† Determined in only 27 hypovolemic and nine hypervolemic patients.
TBV = total blood volume; HR = heart rate; TPR = total peripheral resistance; DBP = diastolic blood pressure.

effects of autonomic activity and the other by imposing an exogenous adrenergic stimulation. In summary, cardiac output and stroke volume are both related to CPV, but at any level of the latter cardiac output will be higher or lower depending on the level of cardiac performance and adrenergic drive.

The number of factors, apart from intravascular volume redistribution, that can affect cardiac output means that contrary to what may be assumed, ECF expansion or hypervolemia does not necessarily mean that output will be higher. This was clearly demonstrated by comparing the hemodynamic characteristics of hypo- and hypervolemic essential hypertensives. In a large unselected group of men and women, cardiac output was equal in both groups despite the marked difference in blood volume (Table 4). This conclusion was confirmed in comparisons restricted to men only in order to avoid possible sex differences in level of cardiac output[93] or in factors influencing fluid balance.[38] The main hemodynamic difference between the two groups was the higher CPV in the hypervolemic hypertensive patients (Table 5). Cardiac output and CPV were significantly correlated ($r = 0.768$, $P < 0.01$) among 12 hypovolemic patients, but the correlation coefficient ($r = -0.642$) for the six hypervolemic hypertensive men did not attain statistical significance, possibly because of the small number involved. Cardiac output (CO) for any particular CPV level was lower in hypervolemic than in hypovolemic essential hypertensives (CO/CPV = 4.07 ± 0.215 and 5.13 ± 0.26, respectively; $P < 0.01$).

To the extent that differences in peripheral resistance can be translated into difference in "vasoconstriction," it is apparent that hypervolemia in hypertension cannot be equated with increased output or less vasoconstriction than other types. In this respect, no relationship was found between PRA and TPR in either group, but PRA was usually lower in hypervolemic patients ($P < 0.025$) (Table 5).

Questions about Total Body Autoregulation in Man

Total body autoregulation is an important element in interpretation of the sequence of hemodynamic events in hypertension.[21] The combination of normal output and high resistance with volemic expansion in some hypertensive patients could be related to the development of autoregulation with persistent hypertension. The sequence of hemodynamic events called autoregulation, namely, an increase of output beyond needs of the tissues followed by rise in peripheral resistance and return of output toward normal, has been demonstrated by Coleman et al.[25] in nephrectomized patients and by Ledingham,[3] Guyton and Coleman,[20] and Ferrario[7] and their collaborators in different types of experimental renal hypertension. The question, therefore, is not whether this hemodynamic sequence occurs, but whether it necessarily always develops and what role it plays in essential hypertension. Some clinical observations are difficult to reconcile with the autoregulation hypothesis; these include the persistence of high output in primary aldosteronism[43] as well as in some patients with long-standing

TABLE 5 *Hemodynamic Indices in Men with Established Essential Hypertension*

Group (no.)	Age (yr)	Weight (kg)	TBV (ml/cm)	HR	DBP (mm Hg)	CI (liters/min per M²)	TPR (U/M²)	Cardiopulmonary volume* ml/M²	Cardiopulmonary volume* CPV/TBV	PRA† (ng/ml)
Hypovolemic (32)	43.6	78	26.2	78	106	2.80	47	508	22.0	2.28
SE	2.0	2.1	0.38	2.0	2.4	0.09	2.2	25	1.2	0.35
Hypervolemic (18)	45.5	94.1	33.4	73	123	2.84	53	654	23.0	1.27
SE	1.7	2.6	0.69	3.4	5.1	0.09	2.7	24	1.2	0.21
P	NS	<0.01	<0.001	>0.10	<0.005	NS	>0.05	<0.01	NS	<0.025

TBV = total blood volume; HR = heart rate; TPR = total peripheral resistance; PRA = plasmin renin activity; DBP = diastolic blood pressure; CI = cardiac index.
* Determined in only 12 hypovolemic and six hypervolemic patients.
† Determined only in 26 and 13 patients, respectively.

essential hypertension[94] and the wide spectrum of hemodynamic patterns reported by Onesti et al.[29, 30] in overhydrated anephric patients.

Ibrahim et al.[94] have described patients with essential hypertension of more than 10 years' duration whose cardiac output was still markedly elevated. The same findings of persistent hyperkinetic circulation were encountered in many patients with primary aldosteronism of over 16 years' duration.[43] Although the output levels at the onset of hypertension were naturally not known, the high values found on repeated studies after years of hypertension suggested that "autoregulation" had not developed. This suggestion was reinforced by the sequential studies of Onesti et al.[29, 30] In a series of impressive reports, they demonstrated (1) a marked contrast in hemodynamic responses of anephric patients to salt and water loading between previously hypertensive and previously normotensive subjects and (2) a wide scatter of cardiac output values in association with hypertension.

In bilaterally nephrectomized hypertensive patients, salt and water loading was allowed to develop for about a month. The hemodynamic correlates of the subsequent pressure rise were quite different; whereas some showed the pattern originally described by Coleman et al.,[25] in others the fluid retention and increase in Na_e, body weight, and blood volume were associated with either a rise in output only for the whole period or with an initial increase in peripheral resistance and no change or even a decrease in cardiac output.[31] The variation in hemodynamic response corresponds to our findings in experimental steroid hypertension in dogs.[45] In this new model of hypertension, sustained rises in arterial pressure develop following the administration of metyrapone, 100/kg per day. The pressure increase was due to persistent elevation in cardiac output in five of 10 dogs with no change in TPR during the 8 weeks of study; the other dogs had various degrees of increased peripheral resistance at different times during their follow-up (Figs. 5 and 6). (These hemodynamic changes were not related to the drug itself since Lefer[95, 96] found that oral metyrapone did not alter cardiovascular hemodynamics, while its intravenous administration produced a dose-dependent cardiac depression and unchanged TPR.) More importantly perhaps, Bravo et al.[45] found that the increased output was not necessarily dependent on hypervolemia, as ECF expansion was not always associated with increased plasma volume. Indeed, measurable increases in ECF did not occur in all dogs although all did develop sustained hypertension.

The relationship of ECF expansion to development of hypertension is often described as a series of quantifiable inescapable steps. The experiences of Onesti et al.[29] suggest that hypertension is not necessarily a preordained result of volume expansion. In their experience with six anephric patients who had been normotensive prior to nephrectomy for end-stage renal disease, salt and water loading led to increased Na_e and hypervolemia but not to hypertension. They also refer to similar experiences by Merrill et al.[97] and by Ducrot et al.[98] On the other hand, we have encountered instances of patients with renal hypertension who became persistently hypotensive following bilateral nephrectomy

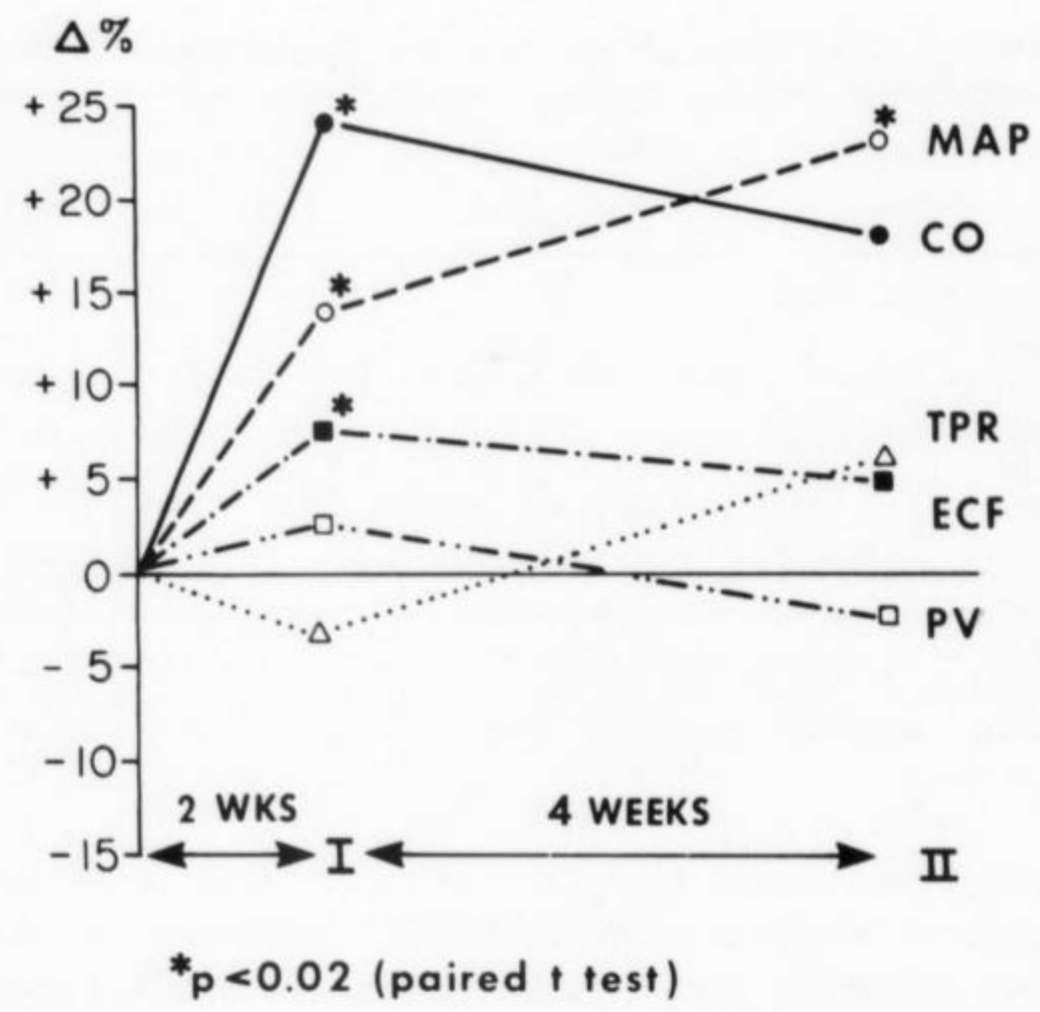

FIGURE 5 *Sequential hemodynamic changes observed in nine dogs given metyrapone, 100 mg/kg per day. Determinations were obtained at weekly intervals. Lack of significance in cardiac output (CO) in period II probably was due to inhomogeneity of hemodynamic patterns (see Fig. 6). MAP = mean arterial pressure; TPR = total peripheral resistance; ECF = extracellular fluid; PV = plasma volume.*

and whose supine arterial pressure remained low despite overt hyperhydration with clinically obvious edema. Although admittedly rare and based on clinical observations rather than on strictly controlled determinations, these examples suggest that the hypertensive responses to fluid overload following bilateral nephrectomy might not always be due to structural arteriolar changes secondary to long-standing hypertension.

In conclusion, the spectrum of volume alterations and of their hemodynamic correlates in hypertension appears too

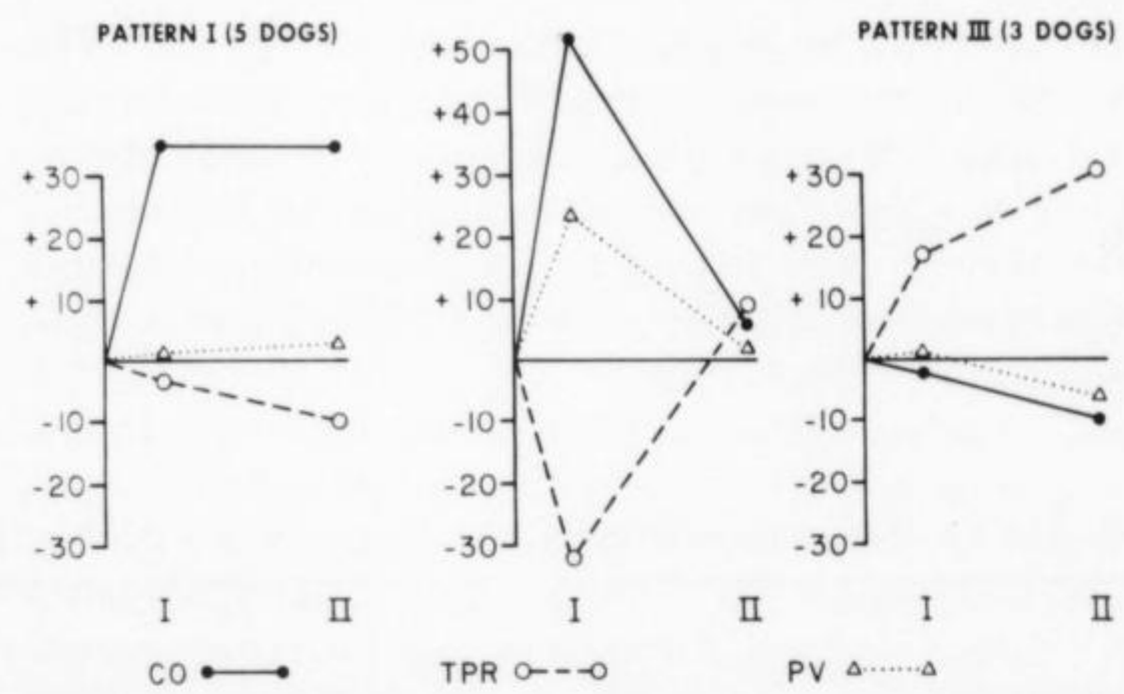

FIGURE 6 *Sequential hemodynamic patterns observed during follow-up of metyrapone-induced hypertension in dogs (based on data by Bravo et al.[45]). Results are expressed as %Δ from control. Points I and II refer to same time intervals as in Figure 5. The hemodynamic patterns depicted in the left hand panel [persistent increase in cardiac output (CO)] and in the right hand panel [rise in resistance (TPR) with no documented change in output] were unchanged during the period of observation; the increase in cardiac output (left hand panel) was significant ($P < 0.001$) at each period. Only in one dog (center panel) was an initial rise in output followed by an increase in total peripheral resistance. PV = plasma volume.*

wide to be forced under single physiological construct. It is clear that there are important plasma volume differences among patients with essential hypertension. These differences are better described in relation to arterial pressure or TPR levels and are associated with borderline changes in ECF and possibly in Na_e. PRA tends to be lower in most hypervolemic hypertensives. However, these volumic differences are not associated with a special hemodynamic pattern. In many instances, the classic pattern of volume overload leading to increased output and eventually increased peripheral resistance, seems to unfold. However, the link between volume expansion and hypertension cannot always be described in these terms.

Luetscher et al.[31] have pointed out convincingly that hemodynamically our biological system can accommodate, without rises in blood pressure, volume alterations of the degree usually seen in hypertensive patients.[9] Neural influences can help regulate the effect of volume expansion at different levels, by influencing ECF distribution (PV/IF ratio), blood volume relocation (CPV/TBV ratio) and cardiac performance (CO/CPV). The fact that hypertension does not necessarily develop with volume overload may indicate the interference of some additional factor in the translation of hypervolemia to hypertension. The importance of genetic factors in that respect was already suggested by the studies of Dahl et al.[97] Recent developments, however, indicate at least three lines of enquiry. The first is the need to reevaluate concepts based on the assumption of "volume expansion" without actual determination of the space investigated. As demonstrated by both human observations[10, 43, 44] and animal studies,[45] these assumptions are not always warranted. Further, extrapolations from ECF to blood volume and vice versa are not always warranted.[34, 44, 76]

The second question which has been hinted at from many studies has come into better focus with the recent sequential hemodynamic observations in anephric patients[30] and in metyrapone-treated dogs.[45] Why do some individuals respond to pressor stimuli with increased peripheral resistance and others with increased cardiac output? This has been reported not only in the abovementioned studies but also in hemodynamic investigations of response to various forms of stress, such as those reported by Brod et al.[100] with mental stress, by Tarazi and Dustan[101] with static exercise, and by Mark et al.[102] in response to salt load. It was also noted by Ferrario and McCubbin[103] in dogs with neurogenic hypertension. Genetic factors undoubtedly play some role, and one may speculate whether predisposition to develop sustained hypertension is more marked in subjects who respond to various stimuli by increasing resistance rather than cardiac output. These differences, however, might not be as clear-cut since in most cases the pressor response could be achieved by alternate routes; when increases in cardiac output were prevented by β-adrenergic blockade, blood pressure still rose to the same degree in response to static exercise,[101] pain,[104] or mental arithmetic.[105]

The third question is whether the links between volume overload and increased peripheral resistance must necessarily go through increased cardiac output and autoregulation. Alterations of the milieu in which arteriolar cells live might be an overriding factor.[106, 107] Another factor is the nervous system; the relationships between ECF expansion and autonomic neural function are multiple. The effect of volume expansion on arterial pressure and cardiac output are deeply modified by autonomic blockade.[108] Deoxycorticosterone hypertension can be prevented by destruction of central adrenergic neurons[109] but not of serotonin-containing nerve terminals.[110] Sodium excess leads to increased norepinephrine turnover and diminished storage in neural granules.[111]

This discussion is not intended to belittle the importance of volume factors in hypertension. The correlation of ECF and plasma volume to blood pressure and TPR has outlined different subgroups of essential hypertension[9] and allowed more specific and effective therapy.[13, 17, 112, 113] In fact, Dustan et al.[15] and Bravo et al.[112] have demonstrated a quantitative relationship between volume contraction and pressure control in volume-responsive hypertensive patients. It is intended to show that the relation between ECF variations and arterial blood pressure is more complex than can be accounted for only by "overloading of the circulation."

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